

CHALK, SKY, WATER, LEAVES, AND THE PROVENANCE OF EXPLANATION

The Cheapness of Because

How teachers can compress science without closing inquiry

A school explanation is necessarily smaller than the science behind it, so the ethical and intellectual question concerns whether the compression carries its provenance, its boundary, and a route back into inquiry. Once explanations cost nothing to obtain, the teacher's work shifts from conveying them towards arguing for what deserves weight, and that shift brings duties with it.

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Full journal references, evidence grades, and classroom protocols included

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The argument in one paragraph

Teachers exercise authority whether or not they claim it, since they choose what enters the lesson, in what order, at what resolution, and under which verb, whether *is*, *causes*, *suggests*, or *evolved for*. When descriptions were scarce that work resembled conveyancing, the careful transfer of a title somebody else had established; now that fluent explanation is free, the scarce goods are discernment and research acumen, and the labour drifts towards advocacy, which is the barrister's craft and carries the barrister's duties. Authority becomes accountable when a compact explanation reveals what kind of answer it is, what evidence warrants it, where it ceases to work, what remains disputed, and which evidence runs against the case being made. The cases in this essay move from chalk to sky, water, green leaves, autumn-red leaves, and the ice on which Feynman's Aunt Minnie slipped. Together they show that physics and chemistry constrain; geology supplies paths, materials, and accidents; biology inherits and modifies; observers classify; and no single "because" is entitled to perform all of those explanatory jobs at once.

Contents

1	The cheapness of <i>because</i>	2
2	The teacher as an accountable authority	2
3	The conveyancer and the advocate	3
4	Chalk: Weinberg's descent and Huxley's return	4
5	Blue sky: a mechanism that still needs an observer	5
5.1	Why not violet? Three repairs in three different places	6
5.2	What a blue sky could be for	7
6	Clear water, coloured water, opaque water	8
7	Green leaves: when mechanism outruns history	9
7.1	The seductive solar graph	9
8	Red leaves: a cost that does not dictate its own reason	10
9	Feynman's magnets, and the ice that dated	11
10	Split the question before answering it	12
11	The pointy rock: teleology arrives early	13
12	Geology and the dignity of accident	13
13	From Kipling to "just-so" science	14
14	Four case files, honestly compressed	15
15	A classroom practice of visible seams	15
15.1	Inquiry needs the vignettes it interrogates	15
15.2	The four-line explanation card	16
15.3	Change the assessment verb	16
15.4	Teach one secure sentence and one live sentence	16
15.5	Make authority inspectable	17
16	A reading route for students and teachers	17
	References	19

1 The cheapness of *because*

Explanations have become easy to obtain. A search box, textbook margin, video, or generative system can produce a polished paragraph before a learner has decided what was being asked. Fluency has become cheap; warrants have not. Misinformation is the obvious hazard and the smaller one, since the subtler danger is a true sentence arriving with the shape and confidence of a complete idea.

“The sky is blue because blue light is scattered more.” “Water is transparent.” “Leaves are green because chlorophyll reflects green.” Each can be useful at the right moment. Each can also create what might be called **epistemic compression with concealed seams**: a small description that suppresses its scale, assumptions, observer, history, and evidential status, then passes as the finished account.

Compression is unavoidable, since a map omitting nothing would be useless and a lesson containing every qualification would never begin. The useful contrast therefore runs between a *sealed* compression and a *provenance-bearing* one, with simplification common to both.

	Sealed compression	Provenance-bearing compression
Claim	Sounds complete; often contains an unmarked “always”, “only”, or “for”.	States the smallest defensible proposition.
Evidence	Hides behind the teacher, textbook, or confident tone.	Names the kind of evidence and, where useful, the source chain.
Boundary	Treats exceptions as nuisances.	Says which scale, system, and conditions the claim covers.
Uncertainty	Either erases it or makes all knowledge sound doubtful.	Separates secure mechanism, bounded generalisation, and open history.
Invitation	Rewards reproduction of the vignette.	Supplies a handle by which the learner can reopen it.

A good short explanation closes one question securely enough to expose the next, giving curiosity traction instead of feeding it until it falls quiet.

2 The teacher as an accountable authority

The legal word *conveyancer* is suggestive. A conveyancer does not create the land being transferred; the work is to establish what is being conveyed, under what title, with which boundaries and encumbrances. Teachers likewise do not manufacture the authority of a spectrometer, field comparison, experiment, or scholarly community. They select and convey claims. The classroom version of a clean chain of title is **provenance**: who claims this, on what evidence, through which transformations, with what reach?

The analogy has limits, but it rescues two truths often pulled apart. First, teachers genuinely exercise authority. They choose examples, omit detail, sequence concepts, translate vocabulary, and decide what counts as an adequate answer. Second, scientific authority is never a personal possession, remaining answerable throughout to methods, evidence, criticism, and the possibility of revision. Ford calls attention to this relation between disciplinary authority and accountability; Allchin similarly argues for teaching the practices and social architecture of “whole science”, not only finished products [Ford, 2008, Allchin, 2011].

Accountability has little to do with a performance of hesitation, since “scientists are not sure” is a poor refrain when the relevant mechanism is secure, and staging every established result as a live debate misleads in the opposite direction. The teacher’s task is more exacting, in that confidence should change when the proposition changes.

Claim type	Typical evidence	Responsible classroom voice
Observation	Calibrated measurement; repeated description	“Under these conditions, we observe. . .”
Mechanism	Intervention, model, physical or chemical constraint	“This occurs through. . .”
Historical route	Geological trace, phylogeny, archival record, converging reconstruction	“The best-supported sequence is. . .”
Current role	Manipulation and measured consequence in a living system	“In this system, the feature can. . .”
Selected function	Comparative and fitness evidence tied to ancestral alternatives	“Evidence suggests selection favoured. . .”
Intended purpose	Evidence of an agent, plan, or interpretation	“This was designed or used in order to. . .”

The distinctions matter because the word *because* can quietly move a class from one row to another. A mechanism becomes a history; a beneficial effect becomes the reason a trait originated; regularity becomes intention.

3 The conveyancer and the advocate

The conveyancing analogy carries a hidden assumption about scarcity, since conveyancing matters in a world where establishing and transferring a title is the difficult part of the transaction. A classroom once sat in that world: the textbook was the deed, the teacher established what it said and passed it on intact, and a pupil who received the description accurately had received something not easily obtained elsewhere. Description now costs nothing. A phone produces a fluent paragraph on chlorophyll, regelation, or ocean colour in the time it takes to ask, and the paragraph is frequently correct.

What remains scarce is discernment, meaning the capacity to tell which of four plausible paragraphs deserves weight, which question each of them is answering, and what would have to be true for one of them to fail. Conveying no longer exhausts the work, and the part that has grown is persuasion, since a learner who cannot yet evaluate a claim must nonetheless be brought to treat some claims as more trustworthy than others. That is the barrister’s function, and the drift towards it is worth calling out, given that the profession’s preferred self-image is the honest broker who merely transmits.

The self-image was always partly a fiction. A teacher who selects the example, fixes the sequence, chooses which caveat to omit, decides what counts as a sufficient answer, and delivers a contested claim in a settled tone has been making a case all along, with the label withheld. The honest response is to accept the advocacy and import the constraints that make advocacy respectable, since denying it loses the constraints along with the label.

Those constraints are unusually well codified. A barrister in England and Wales owes an overriding duty to the court to act with independence in the interests of justice, a duty that displaces

inconsistent obligations to the client; the advocate must not knowingly or recklessly mislead, and must take reasonable steps to ensure that the court has before it all relevant decisions and provisions, which the accompanying guidance makes explicit includes authority adverse to the client’s own case [Bar Standards Board, 2026]. Translated into a classroom, the second obligation is the demanding one. A teacher may argue for a conclusion, and having argued for it owes the class the strongest evidence running the other way, supplied unprompted rather than extracted under questioning. That single duty separates a case from a sales pitch.

The other half of advocacy is examination, which is the half a classroom most needs now. A generated paragraph arrives with the fluency of testimony and none of the exposure, since nobody has asked it what it is in a position to know, how it came to know it, what it would have said had the evidence pointed elsewhere, or whose account it is reproducing. Students can learn to put those questions to a source in the way counsel puts them to a witness, and the questions transfer intact to textbooks, to teachers, and to the students’ own paragraphs.

Role	What it supplies	Characteristic failure
Conveyancer	An accurate title: what the claim says, with its boundaries and encumbrances	Treating transfer as the whole task once transfer has become free
Advocate	A case for what deserves weight, with the adverse authority disclosed	Persuasion that withholds the contrary evidence and calls the omission simplification
Examiner	Pressure applied to a source: what it knows, how it knows, and what would change it	Scepticism practised as a mannerism, applied evenly to strong and weak claims alike

The analogy naturally has limits, since a courtroom supplies an opposing counsel, a tribunal, rules of evidence, and a verdict, while a classroom supplies one voice, a timetable, and an examination board whose mark scheme frequently rewards the sealed vignette. A teacher who ignores the mark scheme damages students in a way that no epistemic virtue repays, and a teacher who obeys it completely reproduces the very thing this essay complains about. The workable compromise is to teach the mark-scheme sentence as what it is, an instrument calibrated for a marker, and then to spend a small protected substantial fraction of the lesson reopening it, which costs minutes rather than schemes of work.

The following objects make those migrations visible.

4 Chalk: Weinberg’s descent and Huxley’s return

Steven Weinberg begins Chapter 2 of *Dreams of a Final Theory* with “On a Piece of Chalk”, [?]. His title recalls T. H. Huxley’s nineteenth-century lecture, which read geological and biological history from chalk [Huxley, 1868].

Why is chalk white? A compact causal descent might run as follows.

Appearance A broad range of visible wavelengths returns to the eye, and the visual system classifies the stimulus as white.



Material optics Weak visible absorption and repeated scattering at grains, pores, cracks, and refractive-index boundaries make the body diffusely bright.



Chemistry Calcium carbonate's electronic structure does not strongly absorb most visible light; impurities can colour the material.



Life and rock Much natural chalk consists of microscopic carbonate remains deposited in ancient seas, buried, compacted, uplifted, quarried, and cut.

Scattering and absorption can be connected to electromagnetism, quantum transitions, particle size, and refractive index [Bohren and Huffman, 1983]. But those laws do not state that this hand specimen had to exist. For that, the explanation turns outward: organisms precipitated carbonate; ecological conditions allowed deposits to accumulate; tectonics and erosion exposed them; human practices placed a fragment in a classroom.

This is reduction: the deeper physics constrains every link, yet a quantum account does not replace constructivist explanations of sedimentology, evolutionary history, or quarrying. Huxley reads upward from the object into history; Weinberg reads downward towards more general law. A complete explanation needs both movements.

A useful compression

Chalk looks white because it absorbs little visible light and multiply scatters many visible wavelengths back towards the observer.

Decompression handle

Why does the microstructure scatter so effectively? Why is the mineral there? Which living processes supplied it? Which geological events made this sample available? The first answer remains true when these questions are opened; it simply stops pretending to be the whole biography.

5 Blue sky: a mechanism that still needs an observer

The canonical school answer is Rayleigh scattering. When particles are much smaller than visible wavelengths, shorter wavelengths are scattered much more strongly; in the idealised regime the dependence is approximately

$$I_{\text{scattered}} \propto \lambda^{-4}.$$

That relation, which Rayleigh derived and then used to account for both the colour and the polarisation of skylight [Strutt, 1871], is indispensable, and it is not yet a colour. Daylight has a spectrum; the atmosphere has gases, aerosols, clouds, density gradients, and paths of different length; the observer looks in a particular direction; and a visual system maps the arriving spectrum onto perceptual categories.

5.1 Why not violet? Three repairs in three different places

The obvious follow-up is the most useful question a class can put to the standard answer. If scattering rises as λ^{-4} , then violet light near 400 nm is scattered about 1.7 times as strongly as blue light near 460 nm, and roughly nine times as strongly as red light near 700 nm, so the sky ought on that reasoning to appear violet. It does not, and the repairs required sit at three different points in the chain, which is the reason no single replacement sentence does the job.

1. **The illuminant is not flat.** The λ^{-4} factor multiplies the solar spectrum reaching the atmosphere, and that spectrum is already falling away at the short-wave end, with ozone and other absorbers removing more of it before scattering occurs. The scattered radiance therefore still climbs towards the violet, and it climbs a good deal less steeply than the scattering law alone suggests.
2. **The medium subtracts as well as scatters.** Ozone absorbs weakly across the orange and red through its Chappuis band, which removes long-wave light from the transmitted beam and deepens the blue that remains. Hulburt calculated in 1953 that the blue of the zenith sky with the Sun at the horizon owes about one third to Rayleigh scattering and two thirds to ozone absorption [Hulburt, 1953], and a recent radiative-transfer study using the CIE colour system reproduces his estimate at the horizon while showing that the ozone contribution is minor for a high Sun and grows with solar zenith angle, viewing zenith angle, and total ozone column [Lange et al., 2023]. The blue of a summer midday and the blue of the blue hour are therefore not the same phenomenon, and the textbook sentence quietly runs them together.
3. **A spectrum has a peak; a colour has coordinates.** This is the load-bearing error in the puzzle. The sky delivers a broad stimulus rather than a spectral line, and the visual system reduces that stimulus to three cone responses before any question of hue arises. Violet light excites the short-wave cones and, through the secondary rise in its sensitivity at short wavelengths, the long-wave cones as well, so a violet-weighted broadband stimulus maps to a desaturated blue; luminous efficiency also falls steeply below 450 nm, and multiple scattering, haze, and ground reflection add broadband light that carries the stimulus further towards white. Smith’s colorimetric treatment carries this through to the chromaticity of the daylight sky, which lands well inside the blue region and well short of the spectrum locus [Smith, 2005].

The third repair is worth dwelling on, since the same confusion returns later in this essay when a solar spectrum is laid beside a chlorophyll absorption curve. Asking which wavelength is most abundant, and asking what colour a mixture appears, are questions about different objects, and a graph answers only the first.

Repair	Where it acts	What it changes about the answer
Source spectrum	Before scattering	The quantity being multiplied by λ^{-4} is already declining in the violet
Ozone absorption	Along the path	Long-wave light is removed; the effect is small overhead and dominant at twilight [Hulburt, 1953, Lange et al., 2023]
Trichromatic encoding	At the observer	A broadband stimulus becomes three numbers, and violet excites two channels rather than one [Smith, 2005]

sky colour = f (stellar spectrum, atmosphere, geometry, observer).

Colour is relational in consequence, which is a different matter from being merely subjective. A spectrometer can report radiance without having a colour experience; a human observer supplies

a species-specific response to that radiance; and another visual system may partition the same photons differently.

Provenance check

Grade the familiar sentence in parts. “Short wavelengths are scattered more” is established, and is the principal mechanism for a high Sun. “Blue is scattered most” is already careless, since violet is scattered more. “Rayleigh scattering explains the blue of the sky” is incomplete for the twilight zenith, where ozone absorption carries most of the effect. And no scattering law, however exact, delivers a colour without an observer to encode the spectrum.

5.2 What a blue sky could be for

The purpose question looks malformed. An atmosphere has no ancestors, no heritable variation, and no differential reproduction, so there is nothing for selection to have acted upon and no selected effect to find. That verdict, however, is about the addressee of the question rather than about the question itself, because only one side of a perceptual relation has a history of selection, and it is not the sky. Eyes have ancestors.

Redirected at the receiver, the question becomes tractable and in places well evidenced. Skylight is the illuminant under which terrestrial vision evolved, and it is highly structured: a polarisation pattern fixed by scattering geometry and stable with respect to haze, a spectral difference between direct sunlight and skylight that gives shade its distinct colour, and a family of daylight spectra ordered by the Sun’s elevation. Several visual systems demonstrably exploit that structure. Desert ants of the genus *Cataglyphis* navigate predominantly by the skylight polarisation pattern, with the sun compass and the polarisation compass running as separate systems fed by different parts of the eye [Wehner and Müller, 2006], and the ecological tradition in vision science treats spectral sensitivities as intelligible in relation to the light available in a habitat [Lythgoe, 1979].

Four questions are therefore hiding inside “what is a blue sky for”, and they have different answers:

Question	Answer	What supports it
Does the sky have a purpose or function of its own?	No	No ancestors, no heritable variation, no selection on atmospheres
Do organisms exploit the sky’s optical structure?	Yes	Behavioural and physiological evidence, as in polarisation-based navigation [Wehner and Müller, 2006]
Was a given visual feature selected because of that structure?	Sometimes	A historical claim needing ancestral alternatives, comparative pattern, and fitness consequences
Does a blue sky mean anything?	To interpreters	Meaning is conferred by minds and cultures, and confers nothing on the atmosphere

The third row is where the just-so risk has migrated, since a spectral sensitivity can be declared “tuned to” the ambient light with very little cost, and the declaration explains any tuning equally well. The history of primate trichromacy is the disciplined version of such a claim, having been argued against opsin genetics, phylogeny, and the measured spectra of the fruits and young leaves at issue, and it remains under argument [Mollon, 1989]. The requirement is the same one this essay applies to red leaves: name the ancestral alternative, and name the observation that would separate

the hypotheses. Exploitation of a feature is never by itself evidence that the feature was made for the user, and the atmosphere in particular was in place long before anything looked at it.

6 Clear water, coloured water, opaque water

Water offers a different trap: treating transparency as a property that is either present or absent. A small clean glass can look colourless while a long path through deep water looks blue. Pure water absorbs visible light weakly and unequally across wavelengths. Careful measurements place a minimum in visible absorption in the blue region, while absorption rises towards the red; over sufficient path length, those small differences accumulate [Pope and Fry, 1997].

The relevant relation is attenuation:

$$I(z, \lambda) = I_0(\lambda)e^{-c(\lambda)z},$$

where z is path length and $c(\lambda)$ combines absorption and scattering out of the measured beam. Increase the distance and a weak spectral preference becomes visible. Add particles or dissolved material and the coefficients change.

Natural waters therefore carry their catchments and ecologies in their optics. Suspended mineral grains scatter; phytoplankton absorb and scatter; coloured dissolved organic matter absorbs strongly at short visible wavelengths; bubbles, bottom reflectance, surface state, and viewing angle contribute. Ocean-colour work has long separated the optical effects of water itself from those of suspended and dissolved constituents [Morel and Prieur, 1977]. Reviews of turbidity likewise show why “muddy” cannot be reduced to one particle count or one visual impression [Davies-Colley and Smith, 2001].

Water seen	Important optical contributors	History carried into the view
Deep blue	Low particle load; preferential loss of longer visible wavelengths; molecular scattering and surface effects	Basin geometry, mixing, biological productivity, and the path through which light travels
Green	Phytoplankton pigments, particles, dissolved matter, and water absorption in varying proportions	Nutrient supply, seasonal growth, food webs, circulation, and catchment inputs
Turquoise glacial lake	Fine mineral “flour” strongly redistributes light	Bedrock mineralogy, glacial abrasion, meltwater routes, and particle settling
Tea-brown lake	Coloured dissolved organic molecules absorb blue light	Soils, vegetation, wetlands, rainfall, microbial processing, and atmospheric deposition [Monteith et al., 2007]
Opaque or turbid water	High suspended sediment, dense plankton, bubbles, or mixed loads	Erosion, floods, land use, resuspension, blooms, and basin disturbance

Is water transparent *so that* life can photosynthesise? The visible transmission window is physically real and biologically consequential. Life has evolved within it and altered aquatic optics in return. But exploitation does not reverse the arrow of explanation. The molecular structure of water was not selected by aquatic organisms. A lake’s present clarity can, however, be causally affected by organisms, geology, weather, and human activity. “Physics below, biology above” is already too static: the levels feed back.

A useful compression

Pure water transmits visible light relatively well, but not equally at every wavelength; natural waters add particles and dissolved substances whose origins are geological, biological, atmospheric, and human.

Decompression handle

Ask for path length, wavelength, angle, constituents, and basin history, since “transparent” names an observation made at a particular scale, with no timeless essence behind it.

7 Green leaves: when mechanism outruns history

Healthy leaves commonly return a larger fraction of middle visible wavelengths to an observer than of blue and red wavelengths. Chlorophylls absorb strongly in blue and red regions; carotenoids add absorption and photoprotection; internal air–cell–water boundaries scatter light; some photons are reflected, some transmitted, and some absorbed. Classic leaf spectra make those components measurable [Gates et al., 1965, Woolley, 1971].

The familiar phrase “chlorophyll reflects green” is therefore a useful mnemonic with two hidden losses. A pigment primarily absorbs; a whole leaf reflects and transmits because pigments sit within structured tissue. And “less strongly absorbed” does not mean “unused”. Green photons penetrate farther before absorption. Under strong white light, they can drive photosynthesis in deeper tissue more effectively at the margin than added red light, because upper chloroplasts may already be saturated [Terashima et al., 2009, Liu and van Iersel, 2021].

Nor is maximum capture by the first surface necessarily the biological optimum. Carbon dioxide supply, enzyme capacity, temperature, nutrient allocation, repair, and photodamage all constrain the use of excitation. In a canopy, a photon captured by a sunlit top leaf cannot reach a shaded leaf below. Models and crop measurements show conditions in which chlorophyll can be reduced with little penalty to canopy photosynthesis, although the outcome depends on architecture and environment [Walker et al., 2018]. Plants also regulate excess excitation through processes including nonphotochemical quenching [Ruban, 2016]. A broad overview is given in [McCulloch-James, 2026c],[McCulloch-James, 2026b]

7.1 The seductive solar graph

It is tempting to compare a modern solar spectrum with a chlorophyll spectrum and announce an evolutionary puzzle: plants reject the most abundant light. But the conclusion changes with what is plotted. The peak of energy per unit wavelength differs from the peak of photon number per unit wavelength and from a spectrum per unit frequency. Atmosphere, water, neighbouring leaves, and tissue reshape the spectrum before a photon reaches a reaction centre [Kiang et al., 2007]. A peak on one graph contains no ancestral population, no competing pigment variant, and no fitness comparison.

Several historical proposals remain scientifically interesting: pigment chemistry may constrain available designs; weaker green absorption may distribute light; reduced fluctuation or overload may matter; ecological spectral partitioning may matter; and an inherited photosystem may be locally improvable but difficult to replace. Quantitative work can show that a proposed optimisation principle reproduces observed antenna spectra [Arp et al., 2020]. Such success supports a candidate organising principle but will never actually film the selection pressures of deep time.

Claim	Status	What follows
Leaves commonly look green because of wavelength-dependent absorption, scattering, and transmission.	Established	Optical spectra and tissue anatomy support the mechanism.
Plants do not use green light.	False	Green light can drive substantial photosynthesis, especially deeper in tissue.
Partial green transmission can distribute excitation through a leaf or canopy.	Strong, bounded	A current benefit is supported under relevant conditions.
Green leaves originally evolved <i>in order to</i> avoid damage or spread light.	Open history	Present usefulness does not uniquely establish the ancestral target of selection.

The optical *how* is secure. Several present roles are strongly supported. The singular evolutionary *why* remains unsettled [Nishio, 2000]. That layered answer carries more information than a neat story, because its confidence is attached to particular propositions.

8 Red leaves: a cost that does not dictate its own reason

Autumn makes the distinction sharper. During senescence, chlorophyll is dismantled through regulated pathways while nutrients are remobilised from the leaf [Hörtensteiner, 2006]. Yellow and orange carotenoids often become more visible as chlorophyll declines. In many red-turning species, anthocyanins are newly synthesised and accumulated in vacuoles. That chemistry is well established. Why a plant makes those pigments at a leaf’s end is not settled by identifying them.

Experiments support photoprotective effects in some species and conditions. Anthocyanic tissue can attenuate light, reduce photoinhibition, and in some systems assist continued nutrient recovery [Feild et al., 2001, Hoch et al., 2001, 2003]. But function varies. A within-leaf study of Norway maple found no significant photoinhibition difference between red and yellow senescing sectors and proposed that redness coincided with stress and carbohydrate disposal rather than supplying substantial photoprotection in that system [Mattila and Tyystjärvi, 2023].

Other hypotheses concern herbivores and signalling: autumn colour might affect insects choosing winter hosts, perhaps advertising defence [Archetti, 2000]. Reviews describe photoprotection, coevolution, carbon balance, antioxidant effects, constraint, and other possibilities; some can operate together rather than compete as a single-elimination tournament [Archetti et al., 2009, Hughes et al., 2022, Lev-Yadun, 2022]. See also [McCulloch-James, 2026a]

The easy story runs: red pigment costs energy; costly traits must purchase a benefit; therefore red evolved for whichever benefit appears plausible. Every semicolon conceals an empirical question. What is the net cost in a sugar-rich senescing leaf? Which alternative phenotype was historically available? Does the effect alter survival or reproduction? Does the comparative pattern follow the predicted environments and lineages? Could a present effect be a later co-option?

Question left open

What observation would distinguish “red was selected because it protected nutrient recovery” from “red is a consequence of carbon imbalance that later gained protective effects”? Evidence compatible with both accounts cannot decide between them merely by being repeated.

Green and red leaves thus teach complementary lessons. With green, a secure optical mechanism coexists with an incomplete ancient history. With red, even some present functions vary among lineages and conditions. “Unsettled” must always name the proposition that is unsettled.

9 Feynman's magnets, and the ice that dated

The most quoted defence of everything argued here comes from an interviewer asking Richard Feynman why magnets repel. Feynman declines the question in its given form and explains the refusal: an answer to a why-question has to terminate somewhere the questioner already accepts, so the shape of an acceptable answer depends on the framework in force. His illustration is domestic. Aunt Minnie is in hospital because she went out, slipped on the ice, and broke her hip, which satisfies everyone in the room and would satisfy nobody from another planet who did not already know why a broken hip leads to a hospital [Feynman, 1983].

He then does what an inquiring class should do, which is to push on the link everyone had accepted. Why did she slip? Ice is slippery, and everybody knows that. Why is ice slippery? At that point Feynman supplies the account he had to hand, that standing on ice momentarily melts it under pressure and produces an instantaneous film of water on which the foot slides, a mechanism available to water because water expands on freezing, so that pressure works to undo the expansion. He then stops and says he is not answering the question so much as demonstrating how difficult a why-question is.

Two features of the passage deserve attention. The first is that Feynman hedged, introducing the pressure-melting account with a parenthetical “they say” that marks it as received rather than established. The hedge is almost always lost in retelling, which is a specimen of exactly the transmission loss this essay is about, since the hedge is the part of a compression that carries its provenance and the part that costs nothing to drop.

The second is that the received account has not survived. Pressure melting is a real effect of the wrong magnitude for the job it was given. The melting point of ordinary ice falls by roughly 0.0074 K for each additional atmosphere, so a skate blade exerting a few hundred atmospheres depresses it by only a couple of degrees, which leaves unexplained both the slipperiness of ice at -20°C and the ordinary experience of slipping while standing still [Rosenberg, 2005]. Two other ingredients carry most of the load. A quasi-liquid layer exists at a free ice surface well below the bulk melting point, a premelting phenomenon Faraday guessed at in the 1850s that surface science has since established and measured [Dash et al., 2006, Weber et al., 2018]; and sliding generates frictional heat, an idea traceable to Bowden and Hughes [Bowden and Hughes, 1939]. Recent measurements complicate the lubrication picture further, since the interfacial water probed during gliding behaves as a viscoelastic film with a viscosity up to two orders of magnitude above that of bulk water, which changes what the film is doing as well as how much of it there is [Canale et al., 2019].

Provenance check

“Pressure melts the ice” should be graded as a real mechanism of insufficient magnitude rather than as a falsehood, and the grading can be made quantitative in a classroom. Ask for the ratio of the pressure-induced depression of the melting point to the actual depression of the air temperature below 0°C . At a few hundred atmospheres and -2°C that ratio approaches unity and the mechanism is live; at -15°C it is small, and the explanation has quietly stopped applying while the sentence carries on sounding identical.

The case earns its place because of who made the error. The physicist most associated with the demand that explanations be understood rather than recited passed on a compressed vignette that was already doubtful when he spoke and is now superseded, in the middle of a passage warning how difficult why-questions are. The lesson is structural rather than personal, since expertise confers no immunity against the sealed compression, which is a property of how knowledge travels through people rather than a defect in the person transmitting it. What protects a class is the habit of asking, of any received sentence, which measurement would have to come out differently for the sentence to fail.

Question left open

Feynman’s regress has no natural stopping point, and he says so; a why-question terminates where the framework in force permits it to. Who chooses that framework in a classroom, and what would a student need in order to challenge the choice?

10 Split the question before answering it

Biologists have offered several maps of “why”. Mayr contrasted proximate and evolutionary causes; Tinbergen distinguished mechanism, ontogeny, function, and phylogeny [Mayr, 1961, Tinbergen, 1963]. David Haig shows that the traditional proximate–ultimate vocabulary has often conflated two different contrasts: immediate versus historical cause, and mechanism versus adaptive function. He recommends clearer questions: *how come?* and *what for?* [Haig, 2013].

Dennett’s contribution is related but distinct. Natural selection can produce design-like organisation and “reasons” without a foresighted reasoner [Dennett, 1995]. That insight allows disciplined function talk; it does not make every ingenious function story true. The following five-way prompt adapts these traditions for classrooms rather than attributing one canonical list to any single author.

Question	Explanatory job	Example
What is observed?	Description	A leaf returns a spectrum perceived by humans as green.
How is it so?	Present mechanism	Pigment absorption and tissue scattering shape the light that escapes.
How did it come to be so?	Development and history	Genes build chloroplasts; lineages inherited and modified older photosystems.
What does it do now?	Current causal role	Green penetration can spread excitation; anthocyanins can screen light in some senescing leaves.
What was it selected for?	Historical function	A specified effect increased ancestral reproductive success relative to available alternatives.
Who meant it?	Intention or later meaning	An agent designed an instrument, or a community interpreted a forest as significant.

The last two rows are easily confused. Selection can preserve a variant *because of* its consequences without any organism intending the outcome. In that bounded biological sense there can be reasons without reasoners. Conscious purpose still requires an intending agent; personal or cultural meaning requires an interpreter. Causal history, selected function, purpose, and meaning are not synonyms.

These distinctions also cross disciplines. Asking what blue sky is *for* may be misplaced; asking what an eye’s spectral sensitivities are *for* may be biologically tractable. Asking why a lake is brown may require both present mechanism and catchment history, with selected function playing no part at all. Grammar cannot decide which form of explanation the world owes us.

11 The pointy rock: teleology arrives early

In Deborah Kelemen’s work, one purpose-based option said that rocks were pointy “so that animals could scratch on them when they got itchy”, while other options set such functions against physical accounts of formation [Kelemen, 1999].

Across the study’s tasks, primary-school children accepted purpose-based explanations of nonliving natural objects more broadly than adults. The result shows a preference expressed under particular tasks and at particular ages, one that instruction and cognitive conditions can influence, which is a good deal narrower than the claims often built upon it about irrational children or an immutable innate teleology. Indeed, later work found that children aged roughly five to eight could learn, retain, and generalise a coherent population-level account of natural selection through a carefully structured picture story [Kelemen et al., 2014].

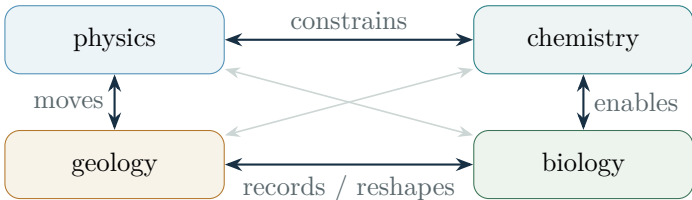
The educational implication is to make the licence for the question *what is it for?* explicit, since banning the question would damage biology, technology, and ordinary practical reasoning alike.

When “for” may be warranted	What must be supplied
Designed artefact	Evidence of an agent, goal, specification, or use
Biological selected function	Heritable alternatives, differential consequences, and historical or comparative support
Current biological role	A demonstrated causal effect, labelled as current rather than ancestral
Personal or cultural meaning	An interpreter or practice that confers the meaning
Rock, sky, or water molecule	No purpose is presumed; ask for mechanism and history unless evidence of use or design enters

Purpose-shaped stories feel satisfying because they compress agent, mechanism, and outcome into one line. That cognitive ease is exactly why the teacher should slow the verb down.

12 Geology and the dignity of accident

It is tempting to draw a hierarchy: physics at the bottom, chemistry above it, biology on top, all resting on geology. The cases here suggest a braid instead.



Chalk is biology sedimented into geology and later interpreted through physics. Life transformed atmospheric chemistry; atmosphere and rock weathering alter waters; waters filter the light in which photosystems evolve; organisms in turn change water colour and sediment. Universal regularities operate at every step, but they do not specify which lawful sequence occurred on Earth.

Feynman made this point without exempting biology or geology from physical law. Historical sciences add the question of how the present arrangement came to be [Feynman et al., 1963]. Gould’s

famous thought experiment of replaying life’s tape emphasised contingency: branching histories depend on earlier events, not only on timeless optima [Gould, 1989]. Evolution experiments make the idea testable; in one long-running bacterial population, access to a key innovation depended on earlier genetic history [Blount et al., 2008].

An **accident** here means an event whose occurrence was not guaranteed by the general laws being invoked, and whose consequences redirected later possibilities, with no violation of law and no absence of causes anywhere in the account. A mutation can obey chemistry yet be undirected with respect to future need. A flood can obey dynamics yet remove one population and spare another. Once history turns, selection can stabilise, refine, or repurpose what contingency supplied.

An Earth-bound accident can therefore be part of a complete reason why without ever amounting to a purpose, and the word *mere* deserves resistance, since contingency is frequently the very information that a historical science is trying to recover.

13 From Kipling to “just-so” science

The phrase belongs to Kipling, whose *Just So Stories for Little Children* appeared in 1902 and offers playful origin tales such as how the camel got his hump [Kipling, 1902]. In science, a “just-so story” has come to mean an attractive post hoc account, especially an adaptive one, whose fit to the known outcome is easier to admire than to test. Gould and Lewontin’s critique of unrestrained adaptationism made the warning famous, although the label and its uses have a more complicated history [Gould and Lewontin, 1979, Hubálek, 2020].

The insult is often used too cheaply. Historical explanations are not defective merely because the past cannot be rerun. Geology, cosmology, and evolutionary biology constrain histories through traces, dates, phylogenies, models, comparative patterns, and experimental analogues. A story becomes scientifically strong when independent evidence makes it difficult to alter at will.

David Deutsch’s useful criterion is that good explanations are *hard to vary while still explaining what they purport to explain* [Deutsch, 2011]. The criterion concerns whether the details do load-bearing work, which is a separate matter from mathematical brevity or emotional austerity. If red leaves can be said to warn insects, hide from insects, attract predators of insects, protect photosystems, dispose of sugar, warm tissue, or merely accompany stress, then “red is useful” is not yet a discriminating explanation. Each candidate must risk different observations.

Hardening move	Question to ask of an appealing story
Specify the contrast	Compared with which available alternative, in which lineage and environment?
Name the mechanism	Through which measurable links does the proposed cause reach the outcome?
Risk a discriminator	What result supports this account over a rival that predicts the same broad pattern?
Fit the chronology	Did the cause, trait, receiver, and environment overlap in time?
Triangulate	Do experiments, comparative data, phylogeny, and environmental reconstruction converge?
State the scope	Is this a local effect, a widespread tendency, or a universal claim?
Name a defeater	What finding would lower confidence rather than be absorbed into the story?

A textbook sentence can be true and still be too easy to vary. Adding a provenance trail and a discriminating question turns an answer-shaped vignette into the beginning of research.

14 Four case files, honestly compressed

The following table models short answers that can survive later decompression. Each is a well-labelled first statement, offered with its boundary attached.

Case	First answer	Boundary	Invitation
Chalk	Weak visible absorption plus multiple scattering makes chalk diffusely white.	Impurities, grain size, moisture, and illumination alter appearance; this is not its geological biography.	How did biological carbonate become this rock and this classroom object?
Blue sky	Small atmospheric molecules scatter shorter visible wavelengths strongly; human vision maps the resulting spectrum to blue.	Aerosols, clouds, solar angle, viewing direction, stellar spectrum, and observer matter.	Which part of the answer changes at sunset, on Mars, or for a different visual system?
Natural water	Water transmits visible wavelengths relatively well but unequally; path length, particles, plankton, and dissolved matter set colour and clarity.	“Clear” is scale- and method-dependent; one constituent rarely explains every lake.	What would absorption, scattering, and catchment measurements each reveal?
Green / red leaves	Pigments and tissue optics set colour. Green photons remain useful; autumn anthocyanins have several supported and disputed roles.	Present mechanism, present benefit, and historical selection require different evidence.	Which claim is secure, which is context-dependent, and which would need a phylogeny or fitness test?

The important phrase in each final column is a question whose answer would change what we are entitled to say, which does considerably more work than an admission that we do not know.

15 A classroom practice of visible seams

Students need recurring cues that knowledge has architecture, which a literature review appended to every sentence would never supply, and four small additions can transform an explanation while leaving it digestible.

15.1 Inquiry needs the vignettes it interrogates

Inquiry-based learning is often proposed as the remedy, and it is a partial one with a failure mode worth stating. An inquiry whose destination is the sanctioned vignette, reached along a worksheet that has removed every wrong turning, rehearses the appearance of discovery while delivering the same sealed sentence at greater cost in lesson time. Discernment also refuses to arrive as a transferable technique, since judging whether a claim about canopy photosynthesis is well warranted requires knowing something about canopies, and the case against minimally guided instruction rests on that dependence: novices lack the schemas that make unguided exploration productive, and cognitive load accounts for much of the rest [Kirschner et al., 2006]. The reply from the inquiry side is equally worth holding, namely that well-designed problem-based and inquiry sequences are heavily scaffolded and aim at epistemic practices alongside content [Hmelo-Silver et al., 2007].

Both sides are right about different things, and the practical resolution is unglamorous. Compressed vignettes are the necessary feed, since a student with no stock of secure mechanisms has nothing to exercise discernment upon. What converts the feed into acumen is a small and repeatable addition, in which each vignette arrives with the question of what would separate it from its nearest rival, and in which a lesson occasionally spends ten minutes discovering that the textbook sentence has a boundary. Research acumen grows out of repeated exposure to that gap, slowly, in students who already know a good deal.

15.2 The four-line explanation card

CLAIM What is the smallest exact statement?

WARRANT What observation, experiment, comparison, or model supports it?

BOUNDARY Where, when, or at what scale might it fail?

OPENING What next question does the answer make possible?

The card can sit beside a diagram, a practical, a paragraph, or an exam answer, modelled by the teacher at first and later completed or challenged by the students, with calibrated assertion as the aim and ritual caveating as the failure mode to be avoided.

15.3 Change the assessment verb

Questions such as “State why leaves are green” reward the shortest authorised vignette. Small changes demand epistemic work:

- Give the best one-sentence mechanism, then identify one fact it leaves unexplained.
- Classify each clause as observation, mechanism, current role, history, or purpose.
- Compare two sources: which makes the narrower claim, and which supplies the stronger warrant?
- Name one observation compatible with both hypotheses and one that would separate them.
- Rewrite an overconfident “because” so its scope and confidence are visible.

These moves align with work on epistemic cognition: learners need to judge aims, sources, methods, and standards, none of which the refrain that science is “tentative” ever requires of them [Chinn et al., 2011]. They also answer a practical problem in online reasoning, where fluent presentation and institutional appearance can substitute for source evaluation [McGrew et al., 2018].

15.4 Teach one secure sentence and one live sentence

For every complex case, the teacher can pair:

Secure: what the strongest converging evidence establishes.

Live: what researchers are still distinguishing, and what evidence would matter.

For green leaves: *secure*, green light is less strongly absorbed than much red and blue light, yet can drive photosynthesis deeper in tissue. *Live*, which mixture of constraint, light distribution, protection, and historical contingency best explains the inherited pigment complement. For red leaves: *secure*, many species synthesise anthocyanins during senescence. *Live*, how much photoprotection, carbon balance, herbivore interaction, and lineage history each contributes.

This pairing prevents two symmetrical mistakes: treating the frontier as if nothing were known, and treating a settled lower-level mechanism as if every historical question had been answered.

15.5 Make authority inspectable

A source chain can be short:

classroom claim $\longrightarrow$ review or synthesis $\longrightarrow$ original data.

Students need not follow every chain every time. They should know that the chain exists, that different links do different work, and that the teacher is willing to expose it. Accountable authority says, in effect: *trust this claim in proportion to these practices, within these boundaries; here is how you could check or extend it.*

16 A reading route for students and teachers

The route below moves from accessible framing to research papers. For technical articles, reading the abstract, figures, captions, and discussion can be more valuable than decoding every line.

1. **Start with an ordinary object.** Read Weinberg’s “On a Piece of Chalk” [?], then sample Huxley’s earlier lecture [Huxley, 1868]. Mark where each author descends towards law and where each reconstructs history.
2. **Watch a why-question refuse to terminate.** View the magnets interview [Feynman, 1983], then read Rosenberg on ice [Rosenberg, 2005] and the gliding measurements of Canale et al. [2019]. Locate the hedge in Feynman’s account, and say why his mechanism is insufficient rather than false.
3. **Put the observer into the sky.** Read Smith’s account of daylight sky colour [Smith, 2005]. Separate the scattering spectrum from the human colour response.
4. **Turn transparency into a measurement.** Use Pope and Fry’s pure-water spectrum [Pope and Fry, 1997] and Morel and Prieur’s ocean-colour analysis [Morel and Prieur, 1977]. Ask what changes when the path length or constituents change.
5. **Correct the green-light myth.** Compare classic leaf optics [Woolley, 1971] with the green-light experiments of Terashima et al. [2009] and the review by Liu and van Iersel [2021]. Explain why “looks green” does not entail “cannot use green”.
6. **See a mechanism remain historically open.** Read Nishio [2000] on why higher plants are green. List which propositions are measurements and which are evolutionary inferences.
7. **Enter a live biological dispute.** Contrast the photoprotection results of Feild et al. [2001] and Hoch et al. [2003] with the Norway maple study by Mattila and Tyystjärvi [2023], then use Lev-Yadun [2022] to map the wider hypothesis field.
8. **Watch a purpose bias in the laboratory.** Read the design and response items in Kelemen [1999], followed by the teaching intervention in Kelemen et al. [2014]. Ask what the second study prevents us from concluding from the first.
9. **Learn to split the whys.** Read Haig’s short paper [Haig, 2013] beside Tinbergen [Tinbergen, 1963]. Reclassify each “because” in one textbook page.
10. **Make stories risk evidence.** Read Gould and Lewontin’s adaptationism critique [Gould and Lewontin, 1979] alongside Deutsch on hard-to-vary explanations [Deutsch, 2011]. Choose one appealing account and write a result that would discriminate it from a rival.
11. **Weigh the inquiry argument.** Read Kirschner et al. [2006] against Hmelo-Silver et al. [2007], then decide which activities in your own scheme of work supply the guidance the two sides are arguing about.
12. **Inspect science as a knowledge practice.** Use Ford [Ford, 2008] or Allchin [Allchin, 2011] to audit one lesson: where does authority come from, where is accountability visible, and which seams could be reopened?

Coda: authority with visible seams

Why is chalk white? Why is the sky blue? Why can one lake be transparent, another turquoise, another brown, another opaque? Why are leaves green, why do some turn red, and why is ice slippery?

Because photons meet molecules, particles, structures, paths, and observers. Because calcium carbonate accumulated in ancient seas. Because atmospheres and catchments have histories. Because chloroplasts are inherited organisations, not fresh engineering projects. Because selection can retain consequences no organism foresaw. Because lawful accidents close some paths and open others. Because a reason can be a cause or a history without being a purpose; because a function can be real without being intended; because meaning can be made without having manufactured the thing that matters.

That paragraph is broad enough to sound wise and still too compressed to teach from. Its claims have different warrants. The responsible teacher does not renounce authority; the responsible teacher shows its joints.

In an age of easy explanations, factual knowledge remains indispensable. What changes is the educational value of merely displaying it. A learner who can recite the sanctioned vignette may possess an answer while lacking an account. A learner who can state a secure mechanism, locate its boundary, inspect its provenance, distinguish present use from selected history, and formulate a question that separates rival explanations has begun to participate in science.

The purpose of a classroom answer is therefore to refine curiosity rather than to satiate it, and the best compression carries a release catch. Chalk teaches the descent towards law; sky teaches the role of the observer; water teaches scale and catchment; leaves teach function, inheritance, and contingency; ice teaches that even an exemplary explanation, delivered by an exemplary explainer, can quietly date. Together they teach something larger: knowledge is not weakened when its seams are visible. That is where accountability enters, and where inquiry can begin again.

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